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SHIORIcafe Database Roles and Privileges

Activity 4 Progress Documentation

1. Introduction

This document presents the progress of Activity 4: Database Users, Roles, Privileges, and Access Control for the SHIORI Café project.

The activity focuses on inspecting the PostgreSQL database, identifying existing database accounts and group roles, reviewing privileges, and documenting the database access configuration.

2. Project Overview

Project Name: SHIORI Café
Database: grp10_shioricafedb
Schema: shioricafe_schema
DBMS: PostgreSQL 18
Application: PHP 8.2+ with PDO PostgreSQL

SHIORI Café is a café management system that handles customer accounts, reservations, orders, payments, menu information, and café operations.

The database access control activity covers the inspection of database ownership, PostgreSQL roles, schema information, table privileges, sequence privileges, and access control configuration.

3. Database Inspection

The PostgreSQL database was accessed using the postgres account.

The following information was inspected:

- Database existence and current connection.
- Database owner and access control list.
- Existing PostgreSQL login accounts and group roles.
- Role memberships.
- Application schema.
- Application tables and sequences.
- Schema, table, and sequence privileges.
- Default privileges.

The inspection confirmed that:

- The database grp10_shioricafedb exists.
- The database owner is postgres.
- The database datacl is NULL.
- The schema shioricafe_schema was not found.
- No SHIORI-specific login accounts or group roles were found.
- No SHIORI-specific role memberships were found.
- No application tables or sequences were found under the target schema.

4. Role Creation and Configuration

The existing PostgreSQL roles were inspected through the PostgreSQL system catalogs.

The following general group roles were identified:

- app_runtime_grp
- app_admin_grp
- activity4_test_role

These roles are configured as NOLOGIN roles.

The inspection also checked for SHIORI-specific accounts and group roles, including:

- shioricafe_app_login
- shioricafe_readonly_login
- shioricafe_migration_login
- app_runtime_grp
- app_readonly_grp
- app_migration_grp

No SHIORI-specific database accounts or group roles were found.

5. Privilege Assignment

The database privilege configuration was inspected using PostgreSQL system catalogs and privilege-checking functions.

5.1 Database and Schema Privileges

The database owner and database access control list were inspected.

The database grp10_shioricafedb is owned by postgres, and its datacl is NULL.

The schema shioricafe_schema was not found during inspection. Therefore, no schema privileges were listed for the target schema.

5.2 Table Privileges

The application tables under shioricafe_schema were inspected.
No application tables were found under the target schema, so no table-level privileges were listed.

5.3 Sequence Privileges

The application sequences under shioricafe_schema were inspected.
No sequences were found under the target schema, so no sequence privileges were listed.

5.4 Group Roles

The existing PostgreSQL group roles were inspected:

- app_runtime_grp
- app_admin_grp
- activity4_test_role

Role membership information was also checked.
No SHIORI-specific group role memberships were found.

5.5 Default Privileges

The PostgreSQL default privileges for the target schema were inspected.
No SHIORI-specific default privilege configuration was found for shioricafe_schema.

6. Access Control Testing

The access control verification focused on the existing PostgreSQL database configuration.

6.1 Runtime Account

The SHIORI-specific runtime account was checked using the PostgreSQL role catalog.
The account shioricafe_app_login was not found in the inspected database environment.

6.2 Read-Only Account

The account shioricafe_readonly_login was checked using the PostgreSQL role catalog.
No matching SHIORI-specific read-only account was found.

6.3 Schema Owner Account

The account shioricafe_migration_login was checked using the PostgreSQL role catalog.
No matching SHIORI-specific migration account was found.
The database owner was confirmed as postgres.

6.4 Testing Summary

Verification	Result
Database connection	Passed
Database owner verification	Completed
Runtime account inspection	Completed
Read-only account inspection	Completed
Migration account inspection	Completed
Group role inspection	Completed
Role membership inspection	Completed
Schema privilege inspection	Completed
Table privilege inspection	Completed
Sequence privilege inspection	Completed

Default privilege inspection	Completed

6.5 Application Integration Testing

The SHIORI Café application uses PHP with PDO PostgreSQL for database connectivity.
The database configuration and migration files were reviewed to identify the intended database connection and access structure.

7. Issues and Limitations

7.1 PostgreSQL Environment Setup

The database grp10_shioricafedb exists and is accessible through PostgreSQL.

The database is currently owned by postgres. The database ACL is NULL, which means PostgreSQL default privileges apply.

7.2 Role Configuration and Permissions

The inspection identified general PostgreSQL group roles used in the environment.
No SHIORI-specific login accounts, group roles, or memberships were identified.

7.3 Application Integration

The project uses PHP 8.2+ with PDO PostgreSQL.
The application database structure and connection configuration require consistency with the finalized PostgreSQL schema and migrations.

7.4 Default Privileges

The default privilege configuration for the target schema was included in the inspection.
The target schema shioricafe_schema was not found in the current database.

8. Evidence and Repository

The SQL verification script and supporting screenshots are organized under the SHIORI Café evidence folder.

Repository links:

- Repository: https://github.com/123airabels/Car_Rental_Management_System
- SHIORIcafe access-control folder: Car_Rental_Management_System/db-access-control/Group05 at main · 123airabels/Car_Rental_Management_System
- SQL script: Car_Rental_Management_System/db-access-control/Group05/Group5-roles-and-privileges.sql at main · 123airabels/Car_Rental_Management_System
- Evidence folder: Car_Rental_Management_System/db-access-control/Group05/Evidence at main · 123airabels/Car_Rental_Management_System

8.1 Evidence Screenshots

The evidence is organized into the following folders:

- [01] Database Inspection
- [02] Role and Privilege Inspection
- [03] Runtime Account Test

The screenshots document the database information, existing PostgreSQL roles, role memberships, and privilege inspection results.

8.2 GitHub Upload

The Activity 4 SQL verification script and supporting documentation are prepared for repository organization.

9. SQL Scripts Summary

The file SHIORI_Cafe_Activity_4_Verification.sql contains the SQL queries used to inspect the SHIORI Café database configuration.

The script includes the following sections:

1. Database and current connection information.
2. Database owner and access control list.
3. PostgreSQL accounts and group roles.
4. Role memberships.
5. Database privileges.
6. Application schema and ownership.
7. Schema privileges.
8. Application tables.

9. Table privileges.
10. Sequence privileges.
11. Sequence ownership.
12. Default privileges.
13. Runtime readiness verification.
14. Runtime operation testing notes.

The verification script uses SELECT queries to inspect the database configuration without modifying existing database objects or privileges.

10. Verification Results

The SQL verification script was prepared to check the database configuration and access control information.

10.1 Runtime Account Verification

The PostgreSQL role catalog was checked for the SHIORI Café runtime account.
No matching SHIORI-specific runtime login account was found.

10.2 Additional Privilege Verification

The following privilege information was inspected:

- Database privileges.
- Schema privileges.
- Table privileges.
- Sequence privileges.
- Default privileges.

The target schema and its application objects were not found during inspection.

10.3 Additional Security Verification

The inspection also covered:

- Database ownership.
- Existing PostgreSQL roles.
- Role membership.
- Database access control list.
- Separation of SHIORI Café roles from other project accounts.

The existing PostgreSQL roles belonging to other projects were not modified.

11. Issues and Fixes

11.1 Additional Verification Completed

Additional SQL queries were included to verify the following:

- Database ownership and database ACL.
- SHIORI-specific login accounts.
- Existing PostgreSQL group roles.
- SHIORI-specific role memberships.
- Application schema existence.
- Application table and sequence information.
- Schema, table, sequence, and default privileges.

The verification script was organized into one SQL file for easier inspection and documentation.

12. Feedback to the Owning Group

The inspection results provide information about the current SHIORI Café database configuration.

The database exists and is owned by postgres. The application schema, SHIORI-specific accounts, and group roles should be aligned with the project's database documentation and finalized migration files.

The database structure and access permissions should be reviewed together to maintain consistent database organization and security.

13. AI Usage Declaration

AI assistance was used to help organize the SQL verification queries, explain PostgreSQL access control concepts, and prepare the Activity 4 documentation.

The database information and verification results were based on the actual PostgreSQL inspection performed for the SHIORI Café project.

14. Conclusion

The Activity 4 inspection documented the current PostgreSQL database configuration of the SHIORI Café project.

The database existence, ownership, PostgreSQL roles, role memberships, schema information, and available privileges were examined using SQL verification queries.

The results provide a reference for understanding the current database access configuration and maintaining proper database organization and security.